

# Report 24 - ColebrookWhite Model

Madhusudhan Pandey

20 September 2026

## Abstract

This report documents the ColebrookWhite model family in OpenHPLjl. Catalogue status: **prototype**.

## 1 Updated Engineering Context

Implicit Colebrook-White turbulent correlation. It belongs to the Friction models domain.

## 2 Generalized Concept Model

The model preserves the domain connector interface while allowing internal fidelity to evolve independently.

## 3 Alternative Models Available in Literature

This family sits on the common fidelity ladder: boundary/algebraic, lumped dynamic, nonlinear physics-based, and distributed/high-order.

## 4 Governing Mathematics

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left( \frac{\epsilon}{3.7D} + \frac{2.51}{Re\sqrt{f}} \right)$$

The production model may add states, constitutive relations, lookup maps, or spatial discretization.

## 5 OpenHPL / Modelica Interpretation

Report 9 maps this family to the closest OpenHPL, Modelica Standard Library, or OpenIPSL analogue.

## 6 Julia / ModelingToolkit Implementation

Source: `src/FrictionModels/ColebrookWhite.jl`. The file is either an implemented component or an explicit Julia source specification.

## 7 Component Connections and Assembly

The model connects through the stable physical connector for its domain and should not expose internal fidelity choices to neighbouring components.

## 8 Numerical Experiment / Validation

Promotion requires an equation-level test, component experiment, limiting-case comparison, structural check, and connected-system test.

## 9 Expected Outputs

Expected outputs are the physical states, connector variables, and conservation or residual quantities needed to verify this family.

## 10 Engineering Interpretation

Integrate with a nonlinear solver.

## Conclusion

ColebrookWhite now has an explicit source/report pair. Promotion to baseline requires validated equations, parameters, tests, and reference behaviour.